Coherence Analysis Plan

Multi-site recordings across sensory preconditioning: Precon 1 vs Precon 4

Draft for discussion | October 27, 2026 | Based on Dickson et al. (2022), eNeuro 9(2): ENEURO.0414-21.2022

1. Overview

This plan follows the approach of Dickson et al. (2022). The aim is to step back and limit the analysis so that it aligns with the initial hypothesis, which was based on that work.

The general idea is a **simple repeated-measures analysis**. Cue pairs are pooled within Precon 1 and within Precon 4 (or the whole session is averaged). The same analyses are then run for θ and γ, matching the strategy of Dickson et al.

Design at a glance

Question	Does coherence between brain regions change from early (Precon 1) to late (Precon 4) preconditioning?
Animals	n = 8, analyzed as a single cohort (no between-subject factor)
Regions	RSC, dHC, ACC, OFC, PER (POR excluded because it was not reached); recorded bilaterally
Session factor	Precon 1 vs Precon 4
Pair factor	10 region pairs (listed below), all weighted equally
Frequency bands	θ and γ, each analyzed separately, as a band average and across frequency bins
Pooling	Cue pairs pooled within each session (or whole-session average)

The 10 region pairs (levels of the Pair factor)

1. RSC-dHC	2. RSC-ACC	3. RSC-OFC	4. RSC-PER	5. dHC-ACC
6. dHC-OFC	7. dHC-PER	8. ACC-OFC	9. ACC-PER	10. OFC-PER

Notation: n (Factor) × m (Factor) lists each factor with its number of levels. DV = dependent variable. f = frequency bin (bins are specific to each band).

2. Statistical model

All analyses use a **generalized estimating equation (GEE)**, as in Dickson et al., with animal ID as the subject (cluster). Session, Pair, Frequency, and Match are within-subject factors.

Settings that apply to all models

Subject variable	Animal ID
Within-subject variable	Session
Working correlation	Exchangeable
Distribution and link	Gamma or Tweedie with a log link, as in the paper (coherence is bounded 0-1 and skewed). Check the residuals.
Estimator	Robust (sandwich) standard errors, ideally with a small-sample correction, because n = 8 is small for GEE.

Exception: matched vs mismatched models

The DV is a Fisher z of a correlation, which can be negative and is roughly normal. Use a **linear (identity) link with a normal distribution**. A gamma or log link would fail on negative values.

Sturdier alternative

A linear mixed model with a random intercept for animal and Kenward-Roger degrees of freedom is often more reliable than GEE with 8 animals. Use it on the raw coherence values or a transformed version (for example, Fisher z of the square root of coherence). If both approaches agree, report the second as a robustness check.

3. Core analyses

Each analysis below is run **separately for θ and for γ**. Analyses A1-A2 use band-averaged coherence. Analyses B1-B2 use coherence at each frequency bin.

A. Band-averaged coherence

A1 Omnibus model

Design	2 (Session: Precon 1 vs Precon 4) × 10 (Pair)
DV	Mean band coherence
Tests	Session, Pair, Session × Pair

A2 Follow-up for each pair

Design	Precon 1 vs Precon 4 within each of the 10 pairs
DV	Mean band coherence

Correction	FDR across the 10 pairs

B. Coherence across frequencies

B1 Omnibus model

Design	2 (Session: Precon 1 vs Precon 4) × 10 (Pair) × f (Frequency bin)
DV	Coherence at each frequency bin
Tests	Session × Frequency, Session × Pair × Frequency

B2 Follow-up for each pair

Design	2 (Session) × f (Frequency bin), run once for each of the 10 pairs
DV	Coherence at each frequency bin
Correction	FDR across the 10 pairs

4. Checks before collapsing

Hemisphere and signal amplitude need to be checked for each band.

C1 Hemisphere check

Design	2 (Session: Precon 1 vs Precon 4) × 10 (Pair) × 2 (Hemisphere relation: ipsilateral vs contralateral)
DV	Mean band coherence
Decision rule	If there is no effect, collapse across hemispheres and proceed with A1-B2.

C2 Signal amplitude check

Design	2 (Session) × 5 (Region) × f (Frequency bin)
DV	Power at each frequency bin
Purpose	Confirms that coherence changes are not amplitude differences.

5. Matched vs mismatched pair analysis

This analysis asks whether coherence features differ for the **matched** pairings (A-B, C-D) compared with the **mismatched** pairings (A-D, C-B).

Pairing	Cue combinations	Relationship
Matched (true pair)	A-B, C-D	First cue → second cue, paired
Mismatched	A-D, C-B	First cue → second cue, not paired

Both matched and mismatched pairings are first cue vs second cue, so **position is matched**. The only difference is whether the cues were paired.

The analysis tests whether the **shape of coherence across frequency** becomes more similar between paired cues, not whether coherence goes up or down. A pairing effect could appear here without any change in mean coherence, and the reverse is also true.

I think that this is actually necessary to address possibility that the other observed changes might be the result of non associative properties like habituation.

D1 Matched vs mismatched model

Design	2 (Session: Precon 1 vs Precon 4) × 2 (Match: matched vs mismatched) × 10 (Region pair)
DV	Fisher z of the spectral correlation
Model	Linear (identity) link, normal distribution (see Section 2)

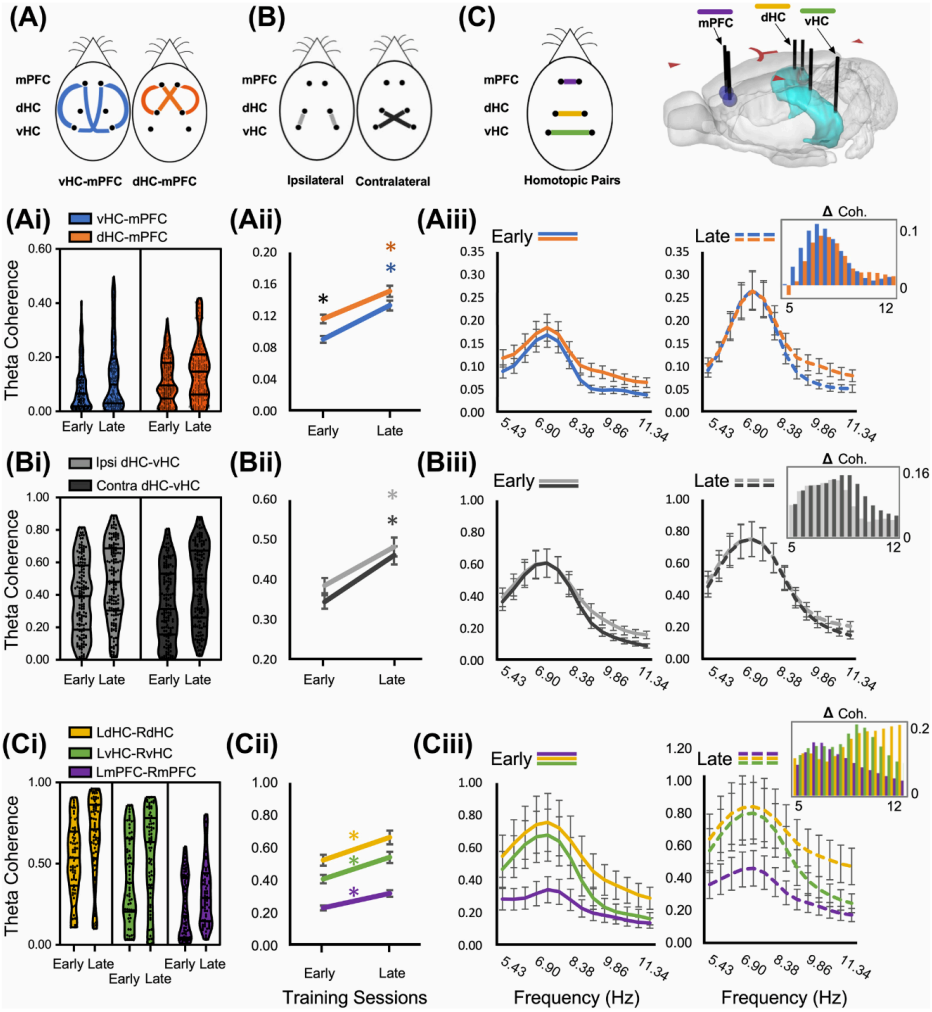
Effects of interest

> **Session × Match** is the key effect. It asks whether the matched-pair link grew relative to the mismatched link across preconditioning.

- **Session × Match × Pair** asks whether that effect differs by region pair.
- **Match main effect** reflects any average similarity difference between matched and mismatched pairings.

6. Planned figures

Figures can follow the format of Dickson et al. (Figure 3, below, panels A-C), with many more lines per panel, or with the region pairs split across separate graphs. Panels i-iii show violin plots, band-averaged coherence by session, and coherence across frequency (Early solid, Late dashed, with a Δ Coh inset).



Reference: Figure 3 from Dickson et al. (2022), eNeuro 9(2): ENEURO.0414-21.2022.

7. Open questions

- **Software.** SPSS has GEE built in but no small-sample variance correction. R (geepack, glmmTMB, lme4 with lmerTest, and emmeans) can do everything above. Which will be used?
- **Collapsing across animals.** Are values collapsed across animals, or calculated per animal and combined later? Proposed: animal is the unit of analysis. Collapse windows (and hemispheres, if C1 is null) within animal, and average across animals only for plots (mean ± SE).
- I think we probably also need to compare FP from day 1 to day 4 to rule out changes in activity that is the result of day or electrodes.